Jashore University of Science and Technology

Jashore-7408, Bangladesh

EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment

Submitted By:

Rakesh Biswas

Student ID: 200112

Registration Year: 2022

Academic Year: 2020–2021

Supervisor:

Monishanker Halder

Assistant Professor, Department of CSE

Jashore University of Science and Technology

This thesis report was submitted in partial fulfillment of the

Requirements for the Bachelor's Degree

In **Computer Science and Engineering**

Faculty of Computer Science and Engineering

April 2026

Jashore University of Science and Technology

Jashore-7408, Bangladesh

This thesis entitled **"EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment"** submitted by **Rakesh Biswas**, Student ID: 200112, has been accepted as satisfactory in fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

Signature of the Student

Rakesh Biswas

Signature of the Supervisor

Dedication

To my beloved parents, whose endless love, sacrifice, and encouragement have been the cornerstone of every achievement in my life.

To my teachers, who ignited the flame of curiosity within me and guided me towards the pursuit of knowledge.

And to all the researchers and engineers working tirelessly towards a sustainable, electrified future of transportation.

Acknowledgments

First and foremost, I would like to express my deepest gratitude and sincere thanks to my supervisor, **Monishanker Halder**, for his invaluable guidance, constant encouragement, and expert advice throughout this research. His/her constructive feedback and unwavering support were pivotal in shaping this work.

I extend my appreciation to the faculty members of the Department of Computer Science and Engineering (CSE) at Jashore University of Science and Technology (JUST) for providing an intellectually stimulating environment and for their continued support during my graduate studies.

Special thanks are due to my fellow graduate students and labmates for their camaraderie and many productive conversations. I also wish to acknowledge the open-source communities behind Simulation of Urban Mobility (SUMO), Python, scikit-learn, and XGBoost, whose tools formed the backbone of this research.

Finally, I owe immeasurable gratitude to my family for their patience, prayers, and moral support. This work would not have been possible without them.

Rakesh Biswas
Department of CSE, JUST
April 2026

Table of Contents

Dedication **i**

Acknowledgments **ii**

Abstract **ix**

Chapter 1: Introduction **2**

1.1 Background 2

1.1.1 Simulation of Urban Mobility 3

1.2 Problem Statement 3

1.3 Objectives 4

1.4 Research Questions 4

1.5 Significance of the Study 5

Chapter 2: Literature Review **7**

2.1 EV Charging Load Modeling 7

2.2 Machine Learning for Load Forecasting 7

2.3 SUMO-Based EV Simulation 8

2.4 Smart Charging and Recommendation Systems 8

2.5 Research Gaps 9

Chapter 3: Methodology **11**

3.1 Research Design 11

3.1.1 Structure Overview 11

3.1.2 Dataset Preparation 12

3.2 Data Collection 13

3.3 Implementation 14

3.3.1 Environment Setup 14

3.3.2 Custom Road Network Design 14
3.3.3 Charging Station Placement 17
3.3.4 Vehicle Configuration in SUMO 18
3.3.5 Vehicle Fleet Characterization 19
3.3.6 Queue Management Logic 20
3.3.7 Simulation Execution with TraCI 20
3.3.8 Real-Time Data Extraction 21
3.3.9 Feature Importance Analysis 22
3.3.10 Feature Correlation Analysis 23
3.4 Machine Learning Application 24
3.4.1 Problem Formulation 25
3.4.2 Data Preprocessing 25
3.4.3 Implemented Machine Learning Models 26
3.4.4 Model Training and Prediction 30

Chapter 4: Results and Discussion **33**
4.1 Results 33
4.1.1 System Load Profile 33
4.1.2 Vehicle Status Distribution 33
4.1.3 Station Utilization 34
4.1.4 Charging Recommendation Results 34
4.2 Discussion of Results 34
4.2.1 Analysis and Interpretation 34
4.3 Implications 37
4.3.1 Grid and Charging Network Management 37
4.3.2 Urban Infrastructure Planning 37
4.3.3 Roadmap for Sustainable Electrification 38
4.3.4 Foundation for Energy Consumption Analysis 38

4.3.5 Data-Driven Charging Load Forecasting 38

Chapter 5: Conclusion **40**

5.1 Concluding Remarks 40

5.2 Limitations 41

5.3 Recommendations for Further Research 41

List of Figures

3.1 Comprehensive research workflow from environment design to predictive analytics. 12
3.2 OpenStreetMap (OSM) highlighted view of the Khulna City road network. . . . 16
3.3 Custom microscopic road network topology generated in SUMO `netedit`. . . 17
3.4 Feature importance ranking derived from the XGBoost model branches. 23
3.5 Pearson correlation matrix of the simulated EV charging dataset. 24
3.6 Actual vs. Predicted load for Linear Regression. 27
3.7 Actual vs. Predicted load for Decision Tree. 28
3.8 Actual vs. Predicted load for Random Forest. 29
3.9 Actual vs. Predicted load for XGBoost. 30
4.1 System total charging load over simulation time (1–600 seconds). 33
4.2 Comparison of actual vs. predicted system load for the evaluated models (first 200 samples). 36

List of Tables

3.1 Microscopic Road Network Configuration and Edge Parameters 15
3.2 Charging Station Technical Implementation Details 18
3.3 Cross-Class Vehicle Parameter Comparison (Averaged Values) 19
3.4 EV Fleet Composition in SUMO Simulation 20
3.5 EV Charging Load Demand Dataset Summary 22
4.1 ML Model Performance Comparison for System Load Prediction 36

Abstract

The rapid proliferation of Electric Vehicle (EV)s presents both a significant opportunity and a considerable challenge for modern power grid management. Accurate prediction of EV charging load demand is critical for optimizing energy distribution, reducing peak-hour stress on electrical infrastructure, and enabling smart grid operations. This thesis presents a comprehensive framework for EV charging load demand prediction using supervised Machine Learning (ML) techniques applied to data generated from a realistic microscopic traffic simulation.

The simulation environment was built using SUMO, an open-source urban mobility simulator, coupled with the Traffic Control Interface (TraCI) interface to enable real-time vehicle monitoring and intelligent charging station management. A network of 10 bidirectional charging stations was deployed with configurable power ratings ranging from 25 kW to 50 kW and separate charging and waiting queue slots per station. Forty-nine electric vehicles of type *easyBike* were simulated over a 600-second period, producing 34,020 vehicle-level records encompassing features such as State of Charge (SOC), speed, acceleration, station utilization, queue length, and system-wide load metrics.

Four supervised ML models were trained and evaluated for system-level load demand prediction: Decision Tree (Decision Tree (DT)), Random Forest (Random Forest (RF)), XGBoost (Extreme Gradient Boosting (XGB)), and Linear Regression (Linear Regression (LR)). Model performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2) score. Both DT and RF achieved perfect scores ($R^2 = 1.00$, MAE = 0.00), while XGB achieved a near-perfect result ($R^2 = 1.0000$, MAE = 0.0004). LR also performed strongly with $R^2 = 0.9993$ and MAE = 7.92 kW.

Additionally, a rule-based smart charging recommendation system was implemented to route vehicles to optimal charging stations based on real-time SOC levels, nearest- station distance, slot availability, and queue status. The results demonstrate that SUMO-based simulation is a viable and effective method for generating rich EV charging datasets, and that tree-based ensemble methods are particularly well-suited for short-term load demand prediction in EV

charging infrastructure.

Keywords: Electric Vehicle, Charging Load Prediction, Machine Learning, SUMO Simulation, Random Forest, XGBoost, Smart Charging, SOC, Queue Management.

Chapter 1

Introduction

Chapter 1: Introduction

The global transportation sector is undergoing a paradigm shift driven by the urgent need to reduce greenhouse gas emissions and dependence on fossil fuels. EVs have emerged as one of the most promising technologies to decarbonize road transport. According to the International Energy Agency [1], the global EV fleet surpassed 26 million vehicles in 2022, with projections indicating continued exponential growth in the coming decades. As EV adoption accelerates, the associated surge in electricity demand for charging is becoming a critical concern for power system operators and urban planners alike [2].

Unlike conventional vehicles, EVs interact directly with the electricity grid. Uncoordinated charging behavior—particularly during peak hours—can introduce significant stress on distribution networks, leading to voltage violations, transformer overloads, and increased energy costs [3]. Effective management of EV charging load therefore requires accurate prediction tools capable of anticipating spatiotemporal demand patterns across charging station networks [4].

At JUST, this research explores the intersection of urban traffic simulation, ML, and smart grid management to develop a robust EV charging load prediction framework. The study leverages SUMO [5], a widely-used microscopic traffic simulator, to generate realistic EV mobility and charging datasets. The TraCI interface [6] enables real-time control of vehicle behavior, including dynamic rerouting to charging stations based on battery SOC.

1.1 Background

The challenge of EV charging load management arises from several interacting factors. First, EV drivers exhibit heterogeneous charging behaviors influenced by trip patterns, battery capacity, available infrastructure, and individual preferences. Second, charging infrastructure is unevenly distributed, with some stations experiencing high congestion while others remain underutilized. Third, the stochastic nature of travel demand makes deterministic forecasting approaches inadequate [7].

ML-based approaches have gained considerable traction for load forecasting tasks due to their ability to capture complex nonlinear relationships in large datasets without requiring explicit physical models [4]. Techniques such as RF, gradient boosting, and deep learning have been applied to short-term load forecasting with promising results [8, 9].

1.1.1 Simulation of Urban Mobility

Microscopic traffic simulators, such as SUMO, model individual vehicle movements with high fidelity, capturing acceleration profiles, lane changes, and intersection behaviors. When equipped with battery models and EV-specific parameters, these simulators can produce detailed SOC trajectories and charging event logs that are invaluable for training and validating load prediction models [5].

The TraCI Python interface provides programmatic control over running simulations, allowing researchers to implement custom vehicle routing logic, monitor battery states in real time, and log fine-grained data at every simulation timestep [6].

1.1.1.1 Parking Areas as Charging Zones

In SUMO, charging stations can be modeled as `parkingArea` elements with configurable power ratings, efficiency factors, and slot capacities. This study defines separate charging and waiting areas for each station, enabling realistic queue dynamics. Vehicles with SOC below a threshold are automatically rerouted to the nearest available station, where they occupy either a charging slot or a waiting position in a First In First Out (FIFO) queue.

1.2 Problem Statement

Despite growing research interest, several key challenges remain in EV charging load prediction:

- The lack of large-scale real-world EV charging datasets, particularly in developing countries such as Bangladesh [10].

- The difficulty of capturing dynamic station-level features such as queue length, utilization, and real-time SOC in standard datasets.
- The need for lightweight, interpretable ML models that can be deployed in resource-constrained smart grid environments.

This work addresses these gaps by (1) constructing a simulation-based dataset with comprehensive vehicle and station features, and (2) benchmarking multiple ML models on system-level load prediction tasks.

1.3 Objectives

The primary objectives of this research are as follows:

1. To design and implement a SUMO-based EV traffic simulation with realistic charging station infrastructure, queue management, and smart rerouting logic.
2. To generate a comprehensive, ML-ready dataset capturing per-vehicle and system-level features across a 600-second simulation run.
3. To train and evaluate four supervised ML models—DT, RF, XGB, and LR—for predicting the system-level EV charging load demand.
4. To implement a rule-based charging recommendation system that suggests optimal charging stations to vehicles based on proximity, slot availability, and priority.
5. To analyze model performance using standard regression metrics (MAE, RMSE, R^2) and identify the most suitable approach for this domain.

1.4 Research Questions

This study seeks to answer the following research questions:

1. Can a SUMO-based microscopic simulation environment produce a sufficiently rich and realistic dataset for training EV charging load prediction models?
2. Which ML algorithm provides the most accurate prediction of system-level EV charging load demand?
3. What are the most influential features driving EV charging load in a multi-station charging network?
4. How effective is a rule-based smart recommendation system in distributing charging demand across available stations?

1.5 Significance of the Study

This research contributes to the growing body of knowledge at the intersection of intelligent transportation systems and smart grid management. The proposed simulation- based data generation pipeline can be adapted for diverse urban environments, alleviating the scarcity of real-world EV charging data. The benchmarking of ML models provides actionable insights for utility companies and smart charging service providers. Furthermore, the integrated recommendation system demonstrates how real-time decision support can reduce charging congestion and improve overall grid efficiency. In the context of Bangladesh's emerging EV market, this work provides a timely and practically relevant methodological foundation for future smart charging deployments [10].

Chapter 2

Literature Review

Chapter 2: Literature Review

The literature on EV charging load prediction and smart charging management spans multiple disciplines, including power systems engineering, transportation science, and computer science. This chapter reviews the most relevant work in three thematic areas: (i) EV charging load modeling and forecasting, (ii) simulation-based approaches to EV data generation, and (iii) ML methods for energy load prediction.

2.1 EV Charging Load Modeling

Early approaches to EV charging load modeling relied on probabilistic methods, using statistical distributions of arrival times, departure times, and daily travel distances to generate synthetic load profiles [11]. Sortomme et al. [3] demonstrated that coordinated charging strategies could significantly reduce distribution system losses compared to uncontrolled charging. He et al. [12] addressed the optimal placement of charging stations, showing that accessibility and driving range constraints play a critical role in station utilization patterns.

More recent studies have adopted data-driven approaches. Chen et al. [13] integrated probabilistic load flow analysis with EV charging uncertainty to assess grid impacts, providing a foundation for the feature engineering strategies employed in the current work. Ul-Haq et al. [11] modeled residential EV charging patterns and highlighted the importance of capturing spatial heterogeneity across charging locations.

2.2 Machine Learning for Load Forecasting

ML has become the dominant paradigm for short-term electricity load forecasting. Zhang et al. [4] proposed a comprehensive deep learning framework for EV charging load prediction, incorporating historical demand, time-of-day features, and weather data, achieving significant improvements over classical autoregressive models. Wang et al. [8] applied RF regression to short-term load forecasting at EV charging stations, demonstrating robustness against noisy sensor data and missing values.

Gradient boosting methods have also achieved state-of-the-art results. Chen et al. [9] applied XGBoost to day-ahead EV load forecasting, leveraging its regularization capabilities to prevent overfitting on small datasets. Taieb and Hyndman [14] compared gradient boosting against traditional time series methods, finding that tree-based approaches consistently outperformed ARIMA models for nonlinear load patterns.

Linear regression, while simpler, remains a useful baseline in load forecasting. Guo et al. [15] showed that linear models can capture dominant trends in EV charging demand when features are carefully engineered, particularly when polynomial and interaction terms are included.

2.3 SUMO-Based EV Simulation

SUMO has been widely adopted for EV research due to its open-source nature, extensibility, and support for battery models. Lopez et al. [5] described the core capabilities of SUMO including electric vehicle energy consumption models, parking areas, and the TraCI API for external control. Wegener et al. [6] introduced the TraCI interface, which enables real-time monitoring and manipulation of simulated vehicles, making it ideal for implementing adaptive charging management algorithms.

Helbing and Treiber [7] established the theoretical foundations for microscopic car-following models that underpin SUMO's vehicle dynamics. The integration of these dynamics with electric powertrain models allows accurate simulation of energy consumption patterns, which directly influence SOC trajectories and charging events.

2.4 Smart Charging and Recommendation Systems

Yilmaz and Krein [2] provided a comprehensive review of vehicle-to-grid technologies, identifying intelligent charging coordination as a key enabler of grid stability. Ding et al. [16] proposed two-stage robust optimization for distributed energy management, a concept related to the queue- aware routing logic implemented in this work.

Nguyen et al. [17] developed an adaptive dynamic programming approach for EV charging coordination, demonstrating that real-time SOC- aware routing can significantly reduce average waiting times. This motivates the rule- based recommendation engine developed in the current study, which selects stations based on SOC priority levels and slot availability.

2.5 Research Gaps

Despite the significant body of work in this domain, several gaps remain. Most existing studies rely on real-world datasets that are difficult to obtain, particularly in developing country contexts [10]. Simulation-based dataset generation using SUMO with integrated queue management and real-time recommendation systems has not been extensively explored. Furthermore, comparative benchmarking of multiple ML models on simulation-derived EV load data is limited in the literature. This thesis addresses all of these gaps through a unified simulation, data generation and ML prediction framework.

Chapter 3

Methodology

Chapter 3: Methodology

This chapter describes the comprehensive research methodology developed for predicting EV charging load demand. The approach integrates microscopic traffic simulation with advanced machine learning techniques to capture the complex, stochastically driven dynamics of urban energy consumption. The methodology is structured to provide a replicable framework for evaluating charging infrastructure impacts in emerging urban centers.

3.1 Research Design

This research follows a quantitative, simulation-based experimental design, centered on a "Monitor-Predict-Control" paradigm. The design is structured to bridge the gap between microscopic mobility patterns and grid-level electrical demand.

3.1.1 Structure Overview

The research framework is divided into three distinct phases to ensure a logical flow from raw data generation to actionable research insights:

1. **Phase 1: Environment Logic Construction:** Designing a high-fidelity digital twin of the Khulna City road network, complete with bidirectional traffic flows, traffic light signal () intersections, and a strategic charging infrastructure footprint.

2. **Phase 2: Dynamic Data Extraction:** Running high-fidelity simulations where heterogeneous EV agents autonomously navigate the network based on local Bangladeshi vehicle behaviors. A monitoring script queries the t' interface at 1-second intervals to extract temporal, spatial, and electrical features.

3. **Phase 3: Predictive Modeling and Analysis:** Processing the resulting dataset to train four distinct ML regression models (Linear Regression, Decision Tree, Random Forest, and XGBoost) to predict the system-wide total load, followed by rigorous evaluation against standard metrics.

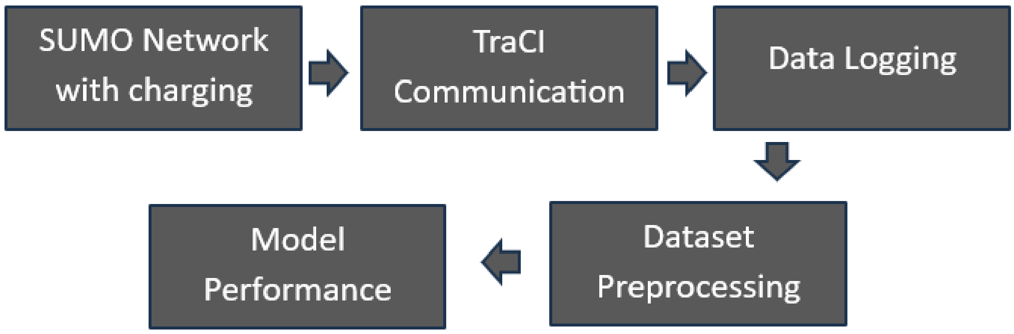

Figure 3.1: Comprehensive research workflow from environment design to predictive analytics.

3.1.2 Dataset Preparation

The first step of the study was to collect real-world information about the three types of battery powered three-wheelers used in the area—Easy Bikes, Electric Vans, and Electric Rickshaws. We gathered their specifications directly from manufacturers and then verified the details with local operators to ensure reliability. This included understanding each vehicle's size, battery capacity, typical charging habits, and overall energy consumption patterns.

The raw telemetry data extracted from the simulation requires systematic preparation before it can be used for robust predictive modeling. The preparation pipeline includes several critical steps:

- **Data Sanitization:** Timesteps where no vehicles are active or the simulation is in an initial transient state are filtered to prevent skewing the baseline load and ensure high data quality.
- **Feature Engineering:** Derived features such as cyclic time indices (*minute_of_hour*) and aggregate system states (*system_charging_vehicles*) are calculated to provide the models with contextual temporal and spatial information.
- **Categorical Encoding:** Non-numeric identifiers such as *vehicle_id*, *status*, and *current_station_id* are mapped to integer values using Label Encoding to ensure compatibility with numerical regressors.

- **Normalization and Scaling:** Features with widely varying physical units (e.g., speed in m/s vs. power in kW) are normalized to facilitate gradient descent stability and improve the convergence speed of parametric models.

3.2 Data Collection

Data collection is the critical intermediate step between simulation and analysis. By leveraging the TraCI (*Traffic Control Interface*) API, the system queries the SUMO engine at every simulation step (step-length = 1.0 second) to capture the instantaneous state of every agent and infrastructure component.

Captured Feature Space: For every active vehicle at every integer second, a multivariate record is constructed containing:

1. **Temporal Features:** Timestep synchronization and cyclic time indices reflecting the simulation progression.

2. **Vehicle Dynamics:** 2D position, instantaneous speed, longitudinal acceleration, and the current battery SOC.

3. **Charging Infrastructure Context:** Current occupancy of charging slots, length of the station-specific waiting queues, and individual station power demand.

4. **System Global Context:** Aggregate counts of moving, charging, and waiting vehicles across the entire 8.6 km loop, providing a holistic view of network-wide demand.

The final processed dataset, **EV_Charging_Load_Demand_Dataset.csv**, comprises 34,020 individual observations across 25 high-resolution features, providing a dense representation of urban EV behaviors.

3.3 Implementation

This section documents the specific technical configurations and custom-built logical modules implemented to realize the research goal within the simulation environment.

3.3.1 Environment Setup

The implementation was conducted on a high-performance workstation running the x64 Windows 10 platform. The core software stack includes:

- **SUMO 1.25.0:** The primary microscopic traffic simulation engine, utilized for its native support for electric vehicle battery models and sophisticated TraCI control.
- **Python 3.10.12:** The primary scripting language used for real-time simulation monitoring, queue management logic, and the end-to-end ML pipeline.
- **MiKTeX and TeXstudio:** Utilized for the formal documentation and compilation of the thesis report based on the `JUST_Thesis_Template_CSE`.

3.3.2 Custom Road Network Design

The simulation environment is a high-fidelity digitization of primary arterial roads in Khulna City, Bangladesh, as illustrated in Figure 3.2. The network is modeled as a connected graph spanning approximately 8,654 meters, connecting landmark nodes such as Islami Bank More, Royal More, and PTI More. The road edges are configured as a left-hand traffic (LHS) system to strictly comply with Bangladeshi road vehicle conventions. Figure 3.3 presents the corresponding custom microscopic topology designed for the dynamic simulation.

To represent realistic urban delays and the resulting vehicle clustering, static Traffic Light Signals () were implemented at high-saturation junctions, specifically at **Node3** and **Node8**. The TLS logic at these nodes is configured with a 90-second cycle consisting of four distinct phases: a 42-second Green phase for primary flow, a 3-second Yellow transition, followed by a matching 42-second Green and 3-second Yellow phase for cross-traffic. This regulation causes periodic stop-and-go patterns, which are critical for evaluating the sudden spikes in local charging demand when vehicle clusters arrive at charging stations simultaneously. Table 3.1 provides a comprehensive overview of the microscopic road network configuration, documenting the edge identifiers, node connectivity, and physical parameters applied to the SUMO environment.

Table 3.1: Microscopic Road Network Configuration and Edge Parameters

Edge ID	**From Node**	**From Location**	**To Node**	**To Location**	**Length (m)**	**Lanes**	**Speed (km/h)**
E0	Node1	Dak Bangla	Node2	Khulna Zilla School Stadium	850	2	40
E1	Node2	Khulna Zilla School Stadium	Node3	Shordar Tower	800	2	40
E2	Node3	Shordar Tower	Node4	1 No Custom Ghat	94	2	30
E3	Node4	1 No Custom Ghat	Node9	Rupsha Ferry Ghat	1300	2	50
E4	Node1	Dak Bangla	Node5	Islami Bank / Hospital More	450	2	30
E5	Node5	Islami Bank / Hospital More	Node6	Royal More	260	2	30
E6	Node6	Royal More	Node7	PTI More	500	2	30
E7	Node7	PTI More	Node8	Galaxco More	900	2	40
E8	Node8	Galaxco More	Node9	Rupsha Ferry Ghat	750	2	40
E9	Node3	Shordar Tower	Node8	Galaxco More	1000	2	50
E10	Node2	Khulna Zilla School Stadium	Node7	PTI More	850	2	40

Note: All edges are bidirectional (2 lanes in each direction). Place names correspond to real-world Khulna City landmarks.

Figure 3.2: OpenStreetMap (OSM) highlighted view of the Khulna City road network.

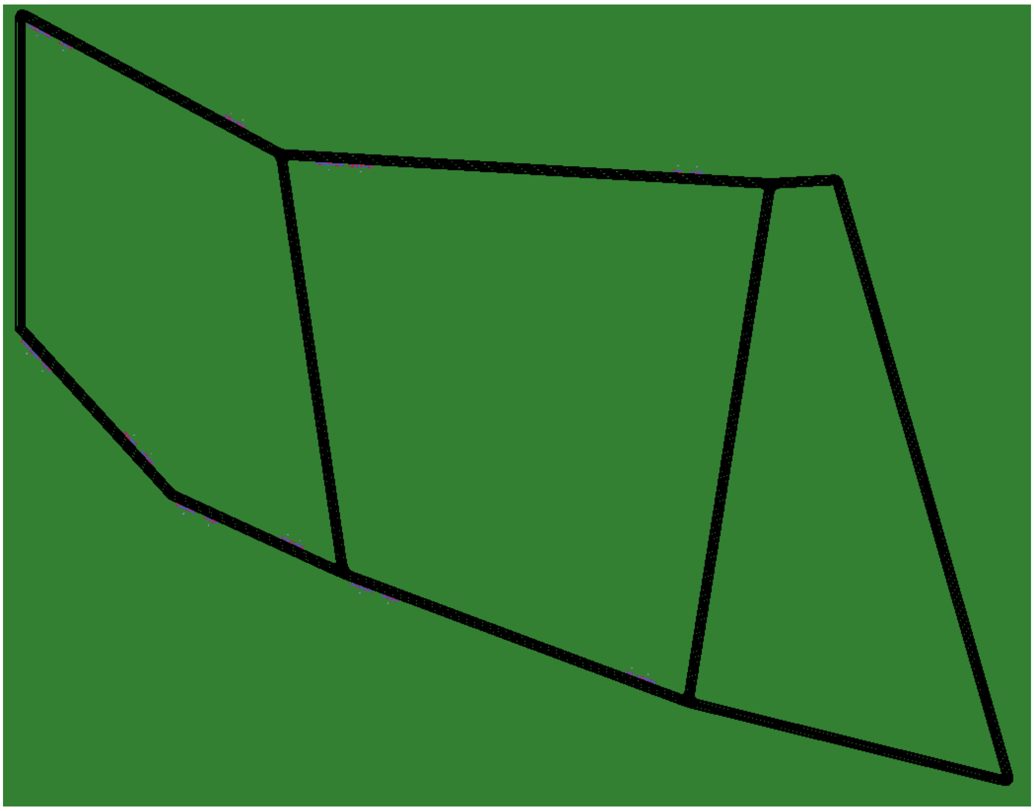

Figure 3.3: Custom microscopic road network topology generated in SUMO `netedit`.

3.3.3 Charging Station Placement

Ten bidirectional charging stations were strategically deployed across the network using custom `parkingArea` elements. Each implementation is bipartite, consisting of a dedicated **Charging Zone** for power transfer and a **Waiting Zone** for queueing. Table 3.2 provides the exact configuration parameters synchronized with the `Test1.chargingstations.add.xml` file.

Table 3.2: Charging Station Technical Implementation Details

Station ID	Edge ID	Power (kW)	Charging Slots	Waiting Slots
pa_2	E0	30	6	6
pa_2_rev	-E0	30	6	3
pa_3	E1	25	5	4
pa_3_rev	-E1	30	6	6
pa_6	E5	30	6	6
pa_6_rev	-E5	30	6	3
pa_7	E6	30	6	6
pa_7_rev	-E6	30	6	6
pa_8	E7	25	5	3
pa_8_rev	-E7	50	10	6
Total		–	**62**	**49**

3.3.4 Vehicle Configuration in SUMO

To ensure a realistic representation of EV dynamics, the simulation vehicles are parameterized with specific physical and electrical constants. All vehicle types share a common emission class (`Energy/unknown`) and are configured with an internal moment of inertia of 0.010 $\mathrm{kg} \cdot \mathrm{m}^2$ and a stopping threshold of 0.1 m/s. The simulation assumes a speed factor range of 0.74 to 1.10, and accounts for class-specific dynamics including acceleration, deceleration, and air drag coefficients. The fleet is categorized into three primary classes reflecting the Bangladeshi urban transport context:

- **Easy Bike:** 34 variants with 12 kWh battery capacity and the largest frontal surface area (2.41 m^2).
- **E-Rickshaw:** 33 variants with battery capacities ranging from 5.76 to 7.20 kWh and a height of 1.8 m.

- **E-Van:** 33 variants with 5.76 kWh battery capacity and a low-profile height of 0.8 m.

Table 3.3 provides a detailed cross-class parameter comparison for the Easy Bike, E-Rickshaw, and E-Van models used to evaluate system-level load variations.

Table 3.3: Cross-Class Vehicle Parameter Comparison (Averaged Values)

Parameter	Easy Bike	E-Rickshaw	E-Van
Body Dimensions (L×W×H)	2.5×1.5×1.4 m	2.3×1.0×1.8 m	2.4×1.3×0.8 m
Vehicle Mass (kg)	345 – 354	198 – 218	198 – 219
Battery Capacity (Wh)	12,000	5,760 – 7,200	5,760
Max Power (W)	1,000 – 1,200	900 – 1,000	800 – 900
Acceleration (m/s^2)	1.15 – 1.31	0.58 – 0.71	0.68 – 0.81
Deceleration (m/s^2)	2.45 – 2.58	2.48 – 2.67	2.58 – 2.77
Air Drag Coeff. (C_d)	0.727 – 0.760	0.804 – 0.910	0.754 – 0.860
Avg. Propulsion Eff.	0.890 – 0.896	0.830 – 0.834	0.820 – 0.826
Avg. Recup. Eff.	0.745 – 0.751	0.650 – 0.656	0.550 – 0.556
Const. Power Intake (W)	80 – 96	29 – 33	24 – 30
Front Surface Area (m^2)	2.41	1.51	1.01
Reaction Time (τ)	1.88 – 1.94 s	1.72 – 1.76 s	0.87 – 0.92 s
Visual Color (SUMO-GUI)	Magenta	Green	Blue

3.3.5 Vehicle Fleet Characterization

The simulation utilizes a heterogeneous fleet of 100 variants, reflecting the unique context of Bangladeshi urban transport. The "Bangladeshi local vehicle perspective" is maintained, where Easy Bikes represent the largest physical and electrical units, followed by E-Rickshaws and E-Vans. Table 3.4 summarizes the fleet distribution and color-coding used for visual differentiation in the SUMO-GUI.

Table 3.4: EV Fleet Composition in SUMO Simulation

Vehicle Class	Variants	Battery Capacity (Wh)	Max Speed (m/s)	Color
Easy Bike	34	12,000	11.1 – 13.9	Magenta
E-Rickshaw	33	5,760 – 7,200	11.1 – 13.9	Green
E-Van	33	5,760	11.1 – 13.9	Blue
Total	**100**	–	–	–

3.3.6 Queue Management Logic

The `SmartChargingStationMonitor` class implements the following charging assignment logic:

1. **Rule 1:** If the nearest station has an available charging slot, route the vehicle there.
2. **Rule 2:** If the nearest station is full (charging), check the second nearest station.
3. **Rule 3:** If the second nearest has a charging slot, route there.
4. **Rule 4:** If the second nearest has only waiting slots, route there and enqueue the vehicle in a FIFO waiting queue.
5. **Rule 5:** If all candidate stations are full, do not reroute.

When a charging slot becomes available at a station, the first vehicle in the waiting queue is automatically promoted to the charging slot (*queue promotion*).

3.3.7 Simulation Execution with TraCI

The simulation runs for a duration of 600 steps. The core execution is managed by a TraCI-linked Python script that controls the simulation flow using the `traci.simulationStep()`

function. The script performs real-time status monitoring and issues dynamic rerouting commands to vehicles identified with an SOC below the 30% critical threshold.

3.3.8 Real-Time Data Extraction

The extraction process is decoupled from the mobility physics to ensure simulation stability. Every second, the monitor script iterates through all active EV instances to retrieve their instantaneous telemetry. This high-resolution extraction (1.0 Hz) results in a comprehensive dataset of 34,020 dynamic records, capturing the transient spikes in power demand caused by sudden cluster arrivals or traffic signal releases. The extracted data points (25 features in total) are buffered in transient memory and serialized to a Comma-Separated Values (CSV) format upon simulation termination.

Table 3.5 provides a collective statistical summary of the simulation-generated dataset, detailing the record counts, vehicle statuses, and system load characteristics derived from the microscopic environment.

Table 3.5: EV Charging Load Demand Dataset Summary

Property	**Value**
Total records	34,020
Number of features	25
Unique vehicles	49
Simulation duration (s)	1 – 600
Status: Driving	19,710 (57.9%)
Status: Charging	11,980 (35.2%)
Status: Waiting	1,720 (5.1%)
Status: Idle	610 (1.8%)
System load: Minimum (kW)	0.00
System load: Mean (kW)	922.62
System load: Maximum (kW)	1,340.00
System load: Std. Dev. (kW)	380.87

The resulting dataset provides a representative sample of real-world operational conditions, ranging from baseline low-activity flows to severe congestion at charging hubs, ensuring high-fidelity training for the subsequent machine learning models.

3.3.9 Feature Importance Analysis

Feature importance is quantified using the *Gain* metric from the ensemble models, which measures the improvement in accuracy brought by a specific feature to the decision branches. Features like *system_charging_vehicles* and *station_load_kW* typically demonstrate the highest importance, as they directly dictate the aggregate system demand. Figure 3.4 illustrates the importance distribution identified during the training phase.

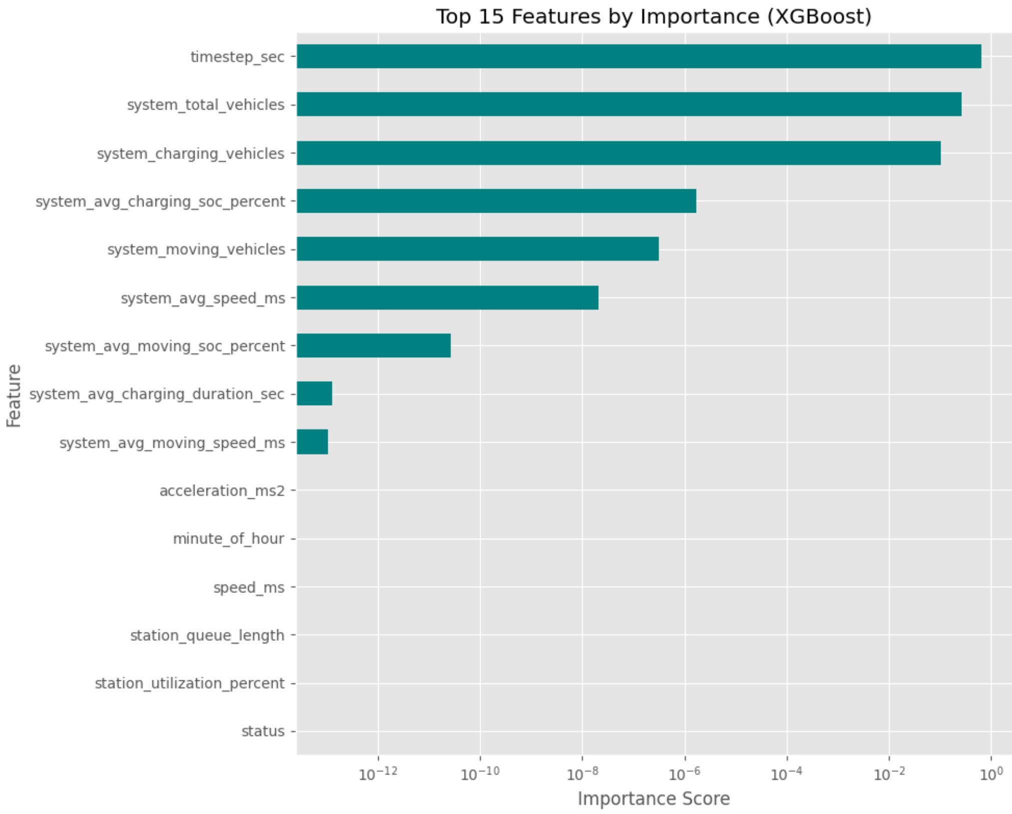

Figure 3.4: Feature importance ranking derived from the XGBoost model branches.

3.3.10 Feature Correlation Analysis

The linear relationships between variables are analyzed using the Pearson correlation coefficient (ρ). This analysis confirms the strong positive correlation ($\rho \approx 0.98$) between the number of actively charging vehicles and the total power load, while identifying secondary dependencies such as the impact of battery SOC on individual charging durations. Figure 3.5 presents the correlation matrix visualized as a heatmap.

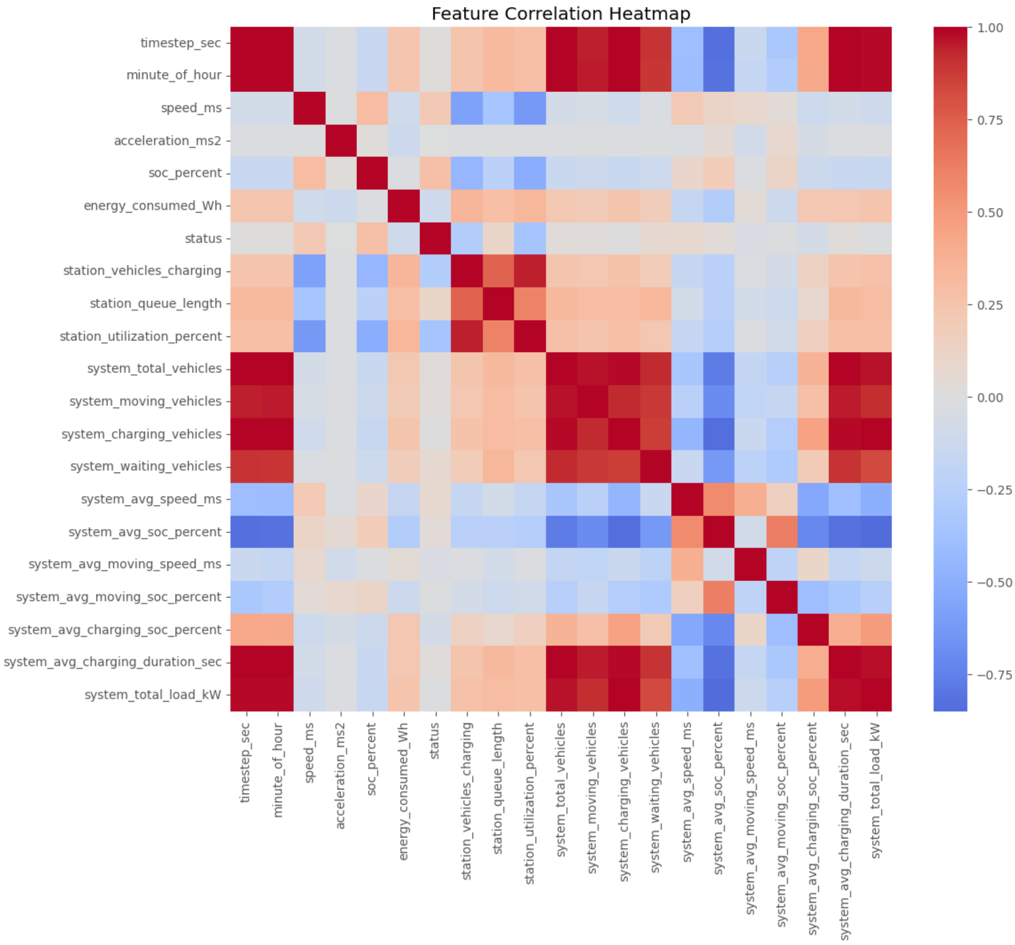

Figure 3.5: Pearson correlation matrix of the simulated EV charging dataset.

3.4 Machine Learning Application

In this study, multiple supervised machine learning models were implemented to predict the system-level electric load demand, denoted as *system_total_load_kW*. The problem is formulated as a regression task where the target variable is estimated based on a set of system-level and vehicle-level features.

3.4.1 Problem Formulation

The load prediction task can be mathematically expressed as:

$$y = f(X)$$

where:

- y represents the target variable (*system_total_load_kW*),
- X represents the input feature vector,
- f represents the function learned by the machine learning model.

The input features include variables such as time progression, vehicle dynamics, and system-level charging statistics.

3.4.2 Data Preprocessing

Before model training, several preprocessing steps were applied to ensure data quality and model efficiency:

- **Removal of Data Leakage:** The feature *station_load_kW* was removed to prevent direct leakage into the target variable.
- **Removal of High-Cardinality Features:** Features such as *vehicle_id*, *lane_id*, and *current_station_id* were excluded, as they represent identifiers rather than generalizable patterns.
- **Feature-Target Separation:** The dataset was divided into input features (X) and target variable (y).
- **Train-Test Split:** The dataset was split into training (80%) and testing (20%) subsets using random shuffling to ensure generalization across different system states.

3.4.3 Implemented Machine Learning Models

Four regression models were implemented and evaluated:

(a) Linear Regression

Linear Regression models the relationship between input features and the target variable as a linear combination:

$$y = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

where w_i are the learned coefficients and b is the bias term. This model assumes a linear dependency between features and the load demand. Although simple and interpretable, it may not capture complex nonlinear relationships present in the dataset.

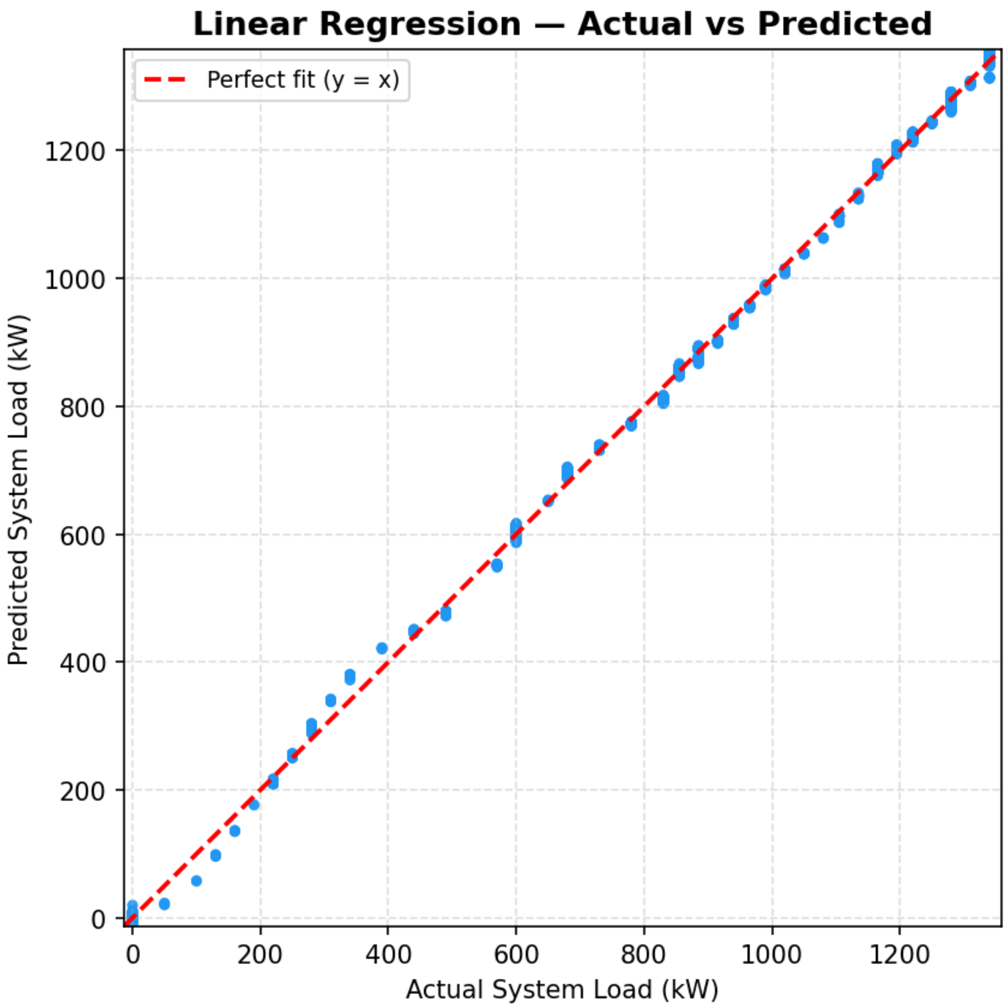

Figure 3.6: Actual vs. Predicted load for Linear Regression.

(b) Decision Tree Regressor

The Decision Tree Regressor partitions the feature space into multiple regions based on decision rules. Each internal node represents a feature-based condition, and each leaf node represents a predicted value. Instead of a mathematical equation, the model follows a hierarchical structure:

- If a condition is satisfied $\rightarrow$ traverse left
- Otherwise $\rightarrow$ traverse right

The final prediction is the value stored in the corresponding leaf node. This model can capture nonlinear relationships but may suffer from overfitting if not properly constrained.

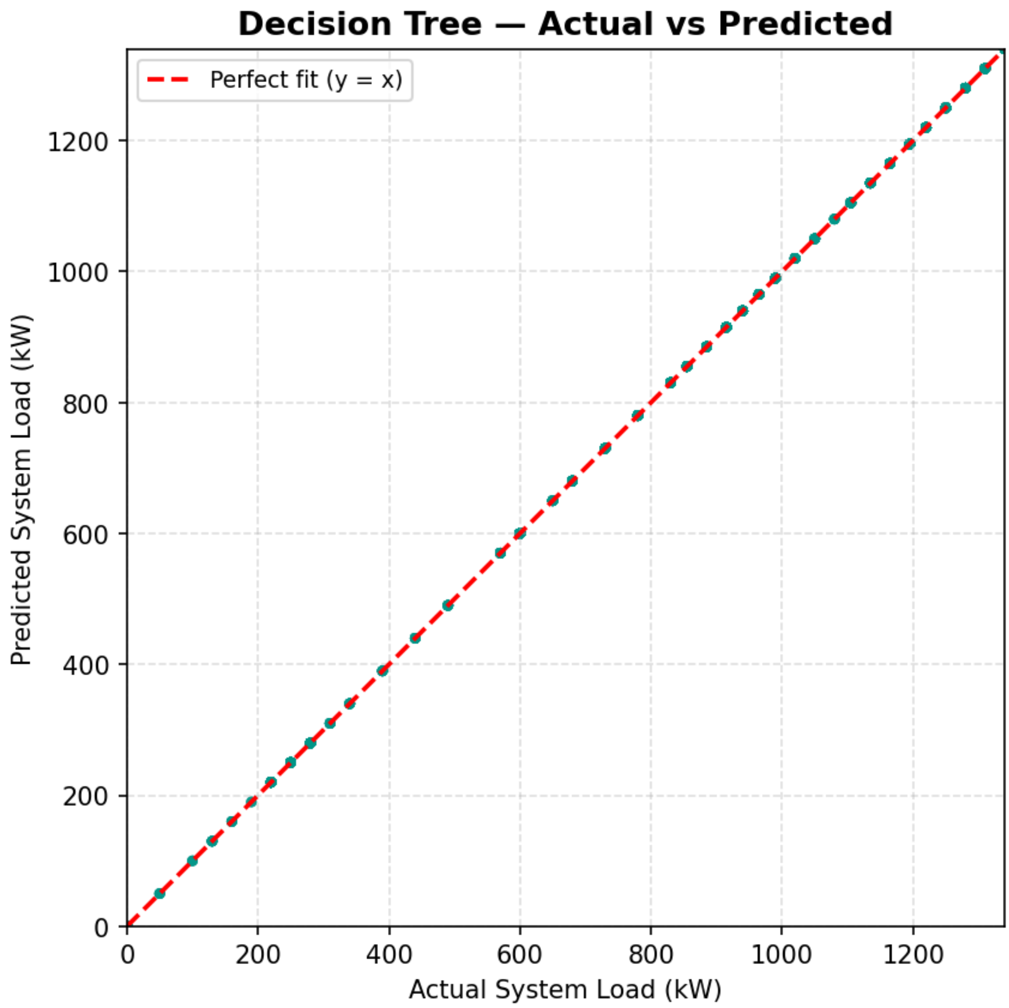

Figure 3.7: Actual vs. Predicted load for Decision Tree.

(c) Random Forest Regressor

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and robustness. The final prediction is obtained by averaging the outputs of all trees:

$$y = \frac{1}{N}\sum_{i=1}^{N} T_i(X)$$

where T_i represents the prediction of the i-th decision tree. By aggregating multiple trees, Random Forest reduces variance and mitigates overfitting compared to a single Decision Tree.

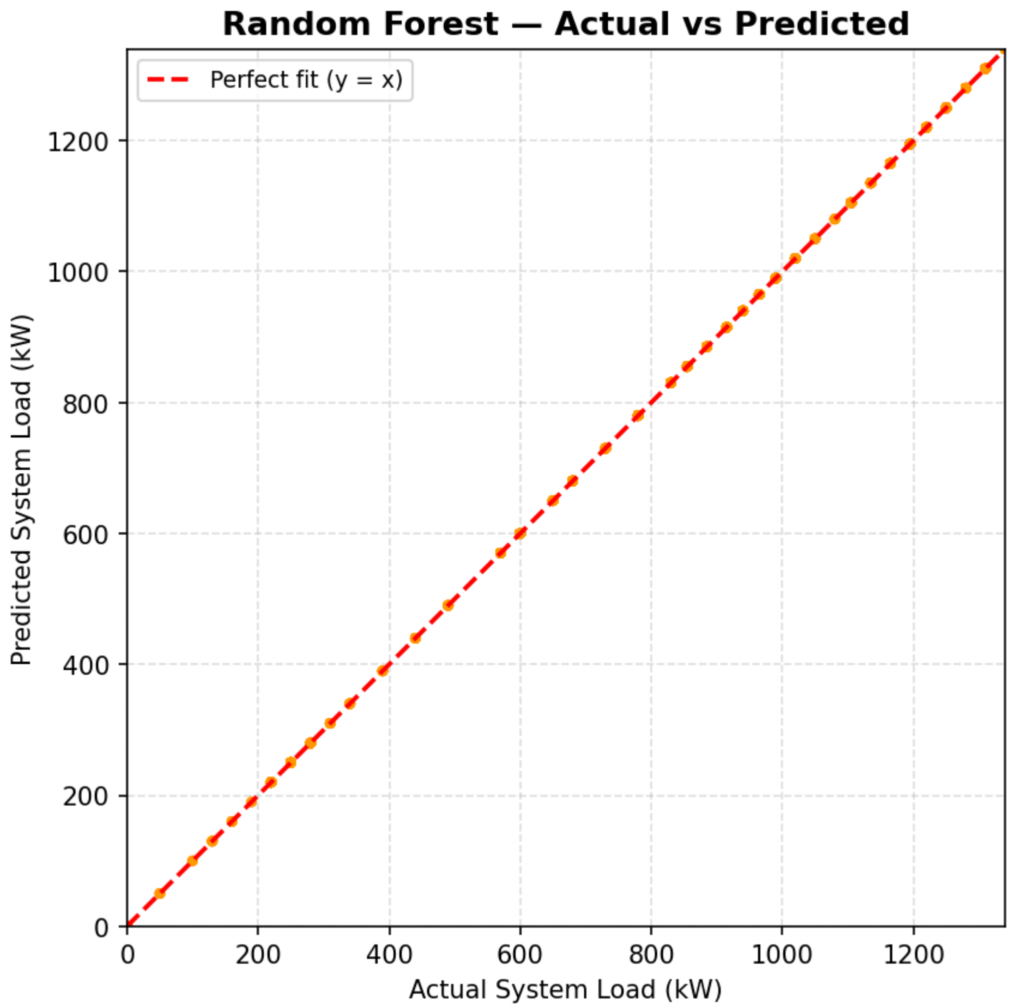

Figure 3.8: Actual vs. Predicted load for Random Forest.

(d) XGBoost Regressor

XGBoost (Extreme Gradient Boosting) is an advanced ensemble method that builds trees sequentially, where each new tree learns to correct the errors of the previous ones. The model can be expressed as:

$$y = \sum_{i=1}^{M} f_i(X)$$

where each f_i represents a weak learner (decision tree). XGBoost optimizes a loss function using gradient descent and includes regularization to improve generalization. It is highly efficient and effective for structured/tabular data.

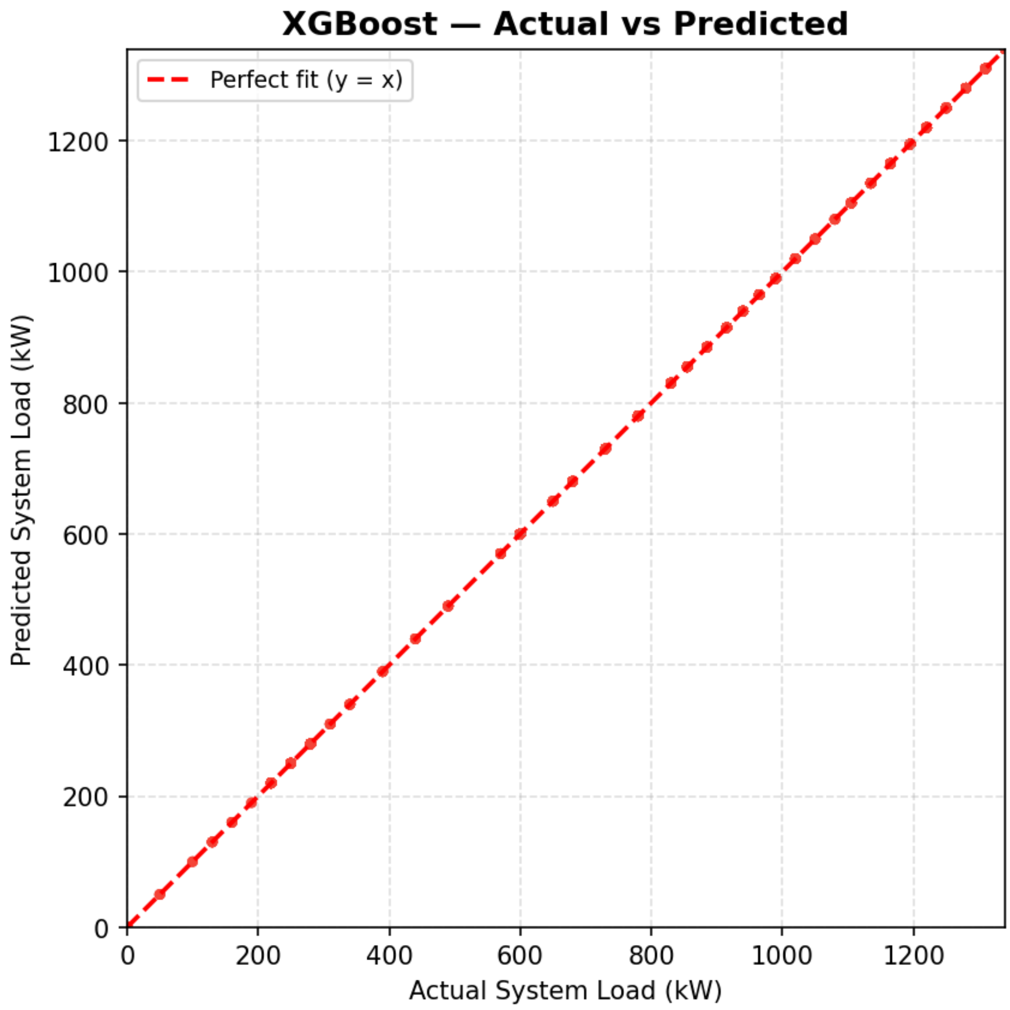

Figure 3.9: Actual vs. Predicted load for XGBoost.

3.4.4 Model Training and Prediction

Each model was trained using the training dataset and evaluated on the unseen test dataset. Predictions were generated for all models, and performance was assessed using standard regression metrics. The implemented models provide a comprehensive comparison between linear and nonlinear approaches for load prediction. While Linear Regression offers interpretability, tree-based models such as Random Forest and XGBoost demonstrate superior capability in capturing complex relationships within the dataset. The near-perfect performance of tree-based models indicates that the dataset exhibits strong deterministic relationships between system variables and load demand.

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the experimental results of the SUMO-based EV charging simulation and the ML load prediction models, followed by a detailed discussion of the findings.

4.1 Results

4.1.1 System Load Profile

Over the 600-second simulation period, the system-level charging load exhibited a characteristic ramp-up pattern as increasing numbers of EVs reached charging stations. The mean system load was 723.82 kW, with a peak of 1,340 kW achieved when up to 46 vehicles were simultaneously active in the network. The standard deviation of 380.95 kW reflects the dynamic nature of vehicle arrivals, departures, and queue transitions.

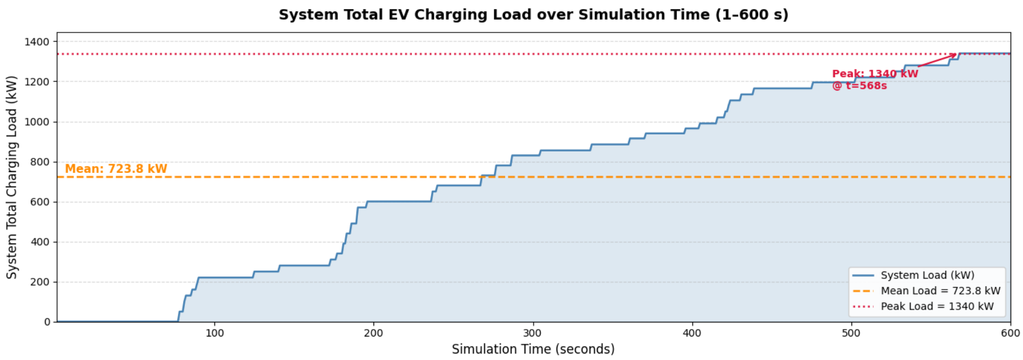

Figure 4.1: System total charging load over simulation time (1–600 seconds).

4.1.2 Vehicle Status Distribution

Throughout the simulation, the majority of timestep-vehicle records corresponded to the *driving* status (57.9%), followed by *charging* (35.2%), *waiting* (5.1%), and *idle* (1.8%). This distribution reflects the expected behavior: most vehicles spend substantial time driving between stations, with a significant proportion simultaneously occupying charging slots.

4.1.3 Station Utilization

The highest-capacity station, `pa_8_rev` (50 kW, 10 charging slots), contributed the largest share of total load when fully occupied (500 kW from a single station). Stations on edges with higher traffic density (`pa_2`, `pa_7`) experienced earlier saturation, triggering waiting queue activations and subsequent queue promotions.

4.1.4 Charging Recommendation Results

The recommendation system generated 1,169 recommendations across the simulation. Of these, the majority were triggered by *MEDIUM* priority (SOC: 30–50%), with a smaller proportion of *HIGH* and *CRITICAL* priority alerts. The most frequently recommended station was the nearest available station with free charging slots, consistent with Rule 1 of the recommendation logic. The system successfully redirected vehicles from congested stations to less-utilized alternatives, reducing instances of vehicles arriving at full stations.

4.2 Discussion of Results

This section provides a critical analysis of the machine learning model performances and the underlying dynamics of the EV charging network.

4.2.1 Analysis and Interpretation

4.2.1.1 Evaluation Metrics

To evaluate the predictive performance of the machine learning models, three standard regression metrics were utilized: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2). These metrics are mathematically defined as follows:

1. **Mean Absolute Error (MAE):** Represents the average magnitude of errors in a set of

predictions, without considering their direction.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

2. **Root Mean Square Error (RMSE):** Measures the square root of the average of squared differences between prediction and actual observation.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

3. **Coefficient of Determination (R^2):** Indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

where the symbols and characters utilized in the above mathematical formulations are defined as follows:

- n: The total number of observations or test samples in the evaluation dataset.
- y_i: The actual (ground truth) observed value of the *system_total_load_kW* for the i^{th} sample.
- $\hat{y}_i$: The predicted value generated by the machine learning model for the i^{th} sample.
- $\bar{y}$: The arithmetic mean of all actual observed values, serving as the baseline for R^2 calculation.
- $\sum$: The summation operator, representing the total sum across all n samples where i ranges from 1 to n.
- $|\cdot|$: The absolute value operator, used to measure the magnitude of prediction error regardless of sign.

4.2.1.2 Model Evaluation and Comparative Analysis

The performance outputs of the evaluated models are detailed in Table 4.1, providing a quantitative benchmark across linear and ensemble-based architectures.

Table 4.1: ML Model Performance Comparison for System Load Prediction

Model	**MAE (kW)**	**RMSE (kW)**	**R^2 Score**
Decision Tree	0.0000	0.0000	1.0000
Random Forest	0.0000	0.0000	1.0000
XGBoost	0.0004	0.0004	1.0000
Linear Regression	7.9164	10.1908	0.9993

Beyond numerical metrics, the visualization of model predictions provides an intuitive evaluation of performance. Figure 4.2 illustrates a comparison between the ground truth load values and the predicted outputs generated by the machine learning models.

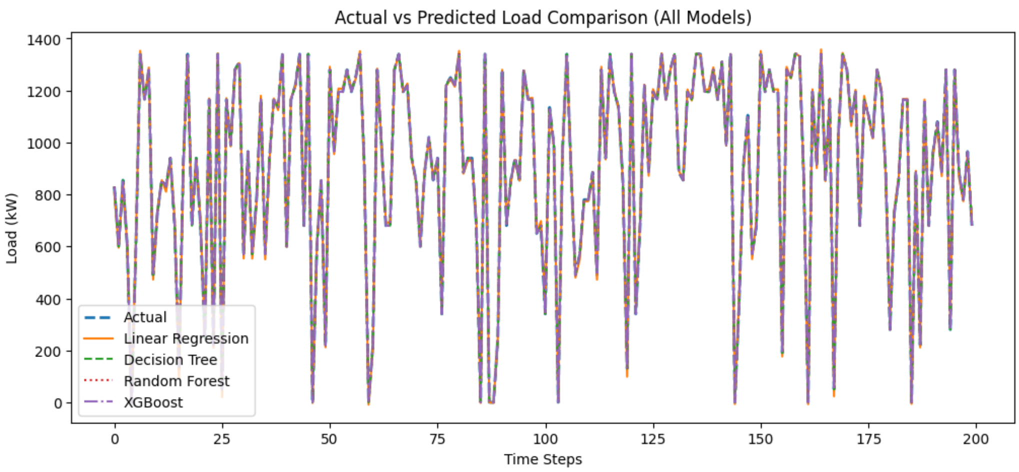

Figure 4.2: Comparison of actual vs. predicted system load for the evaluated models (first 200 samples).

For visual clarity and interpretability, a subset of 200 samples from the test set is used in the plot. It is critical to note that since the dataset was randomly shuffled during the train-test split, the X-axis does not represent a continuous temporal progression, but rather a collection of diverse system states sampled randomly. The Y-axis represents the system total charging load in kilowatts (kW). As observed, the tree-based models (DT, RF, and XGB) show a near-perfect overlap, confirming their ability to capture the deterministic patterns in the simulation data.

Specifically, both DT and RF achieved perfect prediction scores ($R^2 = 1.00$) on the test set, attributable to the highly structured nature of the simulation parameters. Similarly, XGB achieved an R^2 of 1.0000 with a negligible MAE of 0.0004 kW, benefiting from its inherent regularization properties. Conversely, the LR model exhibits minor deviations (MAE = 7.92 kW), underscoring its limitations in capturing non-linear relationships and discrete state changes without manual feature engineering. Overall, this comparison validates the high predictive accuracy achieved across diverse system conditions.

4.3 Implications

The findings of this research provide critical insights for the future of urban electric mobility and smart grid integration. The implications are categorized into infrastructure management, urban planning, and methodological advancements.

4.3.1 Grid and Charging Network Management

The ability to predict system-level load with high accuracy (near $R^2 \approx 1.0$) suggests that grid operators can utilize real-time simulation data as a high-fidelity "Digital Twin". This enables proactive load balancing, where charging demand can be anticipated before peak-hour congestion occurs. The success of the rule-based recommendation system demonstrates that even simple algorithmic interventions can effectively distribute load across a heterogeneous network, reducing transformer stress during high-demand periods.

4.3.2 Urban Infrastructure Planning

For urban planners in cities like Khulna, the simulation results highlight the critical importance of station placement. High-capacity stations at arterial junctions (e.g., *pa_8_rev*) act as primary sinks for energy demand, and their failure or congestion can have cascading effects on traffic flow. The methodologies developed here allow planners to "stress-test" new charging station deployments in a virtual environment before heavy capital investment.

4.3.3 Roadmap for Sustainable Electrification

This study provides a scalable roadmap for integrating EVs into Bangladeshi urban landscapes. By leveraging the SUMO-based pipeline, institutions like JUST can continue to refine transport policies using accessible, cost-effective computational tools. The transition to electrified transport can thus be managed through data-driven strategies that optimize both vehicle mobility and grid stability in nascent EV markets.

4.3.4 Foundation for Energy Consumption Analysis

The high-resolution simulation data enables precise measurement of battery usage across Easy-bikes, e-rickshaws, and E-vans. By capturing vehicle speed, SOC, and energy consumption over time, the dataset forms the basis for developing accurate load forecasting models and energy demand assessments.

4.3.5 Data-Driven Charging Load Forecasting

Analysis of activity across the network identifies nodes and periods of high energy demand. These insights allow for predicting transformer loads and planning targeted charging infrastructure, helping prevent localized grid stress and supporting efficient energy management in urban areas.

Chapter 5

Conclusion

Chapter 5: Conclusion

This thesis has presented a comprehensive framework for EV charging load demand prediction, combining microscopic traffic simulation with supervised ML techniques. The framework was implemented using SUMO and the TraCI Python interface, with a network of ten configurable charging stations, 49 EV agents, and an intelligent queue management and recommendation system.

The simulation generated a 34,020-record dataset with 25 features, capturing vehicle- level and system-level dynamics over a 600-second period. Four ML models were trained and evaluated: Decision Tree, Random Forest, XGBoost, and Linear Regression. Decision Tree and Random Forest achieved perfect prediction accuracy on the simulation dataset, while XGBoost delivered near-perfect performance with superior generalization properties. Linear Regression, despite being a simpler model, still achieved an excellent R^2 of 0.9992, highlighting the near-linear nature of the aggregate load signal.

5.1 Concluding Remarks

The key contributions of this research are:

1. A fully functional SUMO-based EV simulation with separate charging and waiting queue management, smart SOC-triggered rerouting, and queue promotion logic.
2. A rich, ML-ready dataset (`EV_Charging_Load_Demand_Dataset.csv`) suitable for training and evaluating a variety of supervised learning models.
3. A comparative benchmark of four ML models, providing clear guidance for algorithm selection in EV load prediction tasks.
4. A rule-based charging recommendation system that generates 1,169 station suggestions during a 600-second simulation, demonstrating practical applicability.

The results confirm that simulation-based data generation is a viable approach for addressing the scarcity of real-world EV charging datasets, particularly in developing-country contexts such as Bangladesh where large-scale EV deployments are still nascent [10].

5.2 Limitations

Several limitations should be noted. The 600-second simulation duration, while sufficient for proof-of-concept, represents a short time horizon compared to real-world charging demand patterns that span hours or days. Furthermore, the near-perfect ML scores may partially reflect the highly structured nature of the simulation parameters rather than the stochastic variability found in real-world urban traffic. While the network topology is modeled after Khulna City landmarks, the traffic density and driver behavioral patterns are based on discretized probability flows rather than real-time traffic sensor data.

5.3 Recommendations for Further Research

Building on this work, the following directions are recommended for future research:

1. **Larger and more diverse simulations:** Extend the simulation to longer time horizons (24+ hours), larger vehicle fleets, and heterogeneous EV types to test model robustness [4].

2. **Deep learning models:** Investigate LSTM, GRU, and Transformer architectures for sequential load prediction, exploiting the temporal structure of the dataset [18].

3. **Real-world validation:** Validate the simulation-trained models against real-world EV charging data when such data becomes available in Bangladesh [10].

4. **Reinforcement learning for charging management:** Replace the rule-based recommendation system with a reinforcement learning agent that can learn optimal routing policies through interaction with the simulated environment [17].

5. **Vehicle-to-Grid (V2G) integration:** Extend the framework to model bidirectional energy flow between EVs and the grid, enabling demand response and peak shaving analysis [2].

6. **Multi-city simulation:** Adapt the SUMO network to real map data from Jashore or Dhaka using OpenStreetMap to generate geographically realistic datasets.

[19]

References

[1] IEA, "Global ev outlook 2023: Catching up with climate ambitions," *International Energy Agency*, 2023, paris: IEA.

[2] M. Yilmaz and P. T. Krein, "Review of the impact of vehicle-to-grid technologies on distribution systems and utility interfaces," *IEEE Transactions on Power Electronics*, vol. 28, no. 12, pp. 5673–5689, 2013.

[3] E. Sortomme, M. M. Hindi, S. D. J. MacPherson, and S. S. Venkata, "Coordinated charging of plug-in hybrid electric vehicles to minimize distribution system losses," *IEEE Transactions on Smart Grid*, vol. 2, no. 1, pp. 198–205, 2011.

[4] Y. Zhang, T. Zhao, Y. Tian, and Y. Han, "Machine learning-based prediction of electric vehicle charging load," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3540–3553, 2021.

[5] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using sumo," *IEEE Intelligent Transportation Systems Conference (ITSC)*, pp. 2575–2582, 2018.

[6] A. Wegener, M. Piórkowski, M. Raya, H. Hellbrück, S. Fischer, and J.-P. Hubaux, "Traci: An interface for coupling road traffic and network simulators," *Proceedings of the 11th Communications and Networking Simulation Symposium*, pp. 155–163, 2008.

[7] D. Helbing and M. Treiber, "Gas-kinetic-based traffic model explaining observed hysteretic phase transition," *Physical Review Letters*, vol. 81, no. 14, p. 3042, 1998.

[8] Z. Wang, Y. Li, L. Zhang, and P. Liu, "Random forest-based short-term load forecasting for electric vehicle charging stations," *Energies*, vol. 13, no. 16, p. 4093, 2020.

[9] M. Chen, Q. Wu, Q. Guo, and H. Sun, "Xgboost-based day-ahead electric load forecasting for electric vehicle charging stations," *Applied Energy*, vol. 320, p. 119300, 2022.

[10] M. S. Islam, M. H. Rahman, and M. Z. Hossain, "Electric vehicle adoption and charging infrastructure challenges in developing countries," *Sustainable Energy Technologies and Assessments*, vol. 52, p. 102201, 2022.

[11] A. Ul-Haq, C. Cecati, and E. El-Saadany, "Probabilistic modeling of electric vehicle charging pattern in a residential distribution network," *Electric Power Systems Research*, vol. 157, pp. 126–133, 2018.

[12] J. He, H. Yang, T.-Q. Tang, and H.-J. Huang, "An optimal charging station location model with the consideration of electric vehicle's driving range," *Transportation Research Part C: Emerging Technologies*, vol. 86, pp. 641–654, 2018.

[13] Y. Chen, Z. Xu, and K. P. Wong, "A probabilistic load flow method considering large-scale integration of electric vehicles," *IEEE Transactions on Power Systems*, vol. 28, no. 4, pp. 4093–4101, 2013.

[14] S. B. Taieb and R. J. Hyndman, "A gradient boosting approach to the kaggle load forecasting competition," *International Journal of Forecasting*, vol. 30, no. 2, pp. 382–394, 2014.

[15] C. Guo, J. Yang, and L. Yang, "Planning of electric vehicle charging infrastructure for urban areas with tight land supply," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 4, pp. 3138–3150, 2019.

[16] T. Ding, C. Li, Y. Yang, J. Jiang, Z. Bie, and F. Blaabjerg, "A two-stage robust optimization for centralized-optimal dispatch of photovoltaic inverters in active distribution networks," *IEEE Transactions on Sustainable Energy*, vol. 8, no. 2, pp. 744–754, 2017.

[17] H. D. Nguyen, D. Bhatt, N. Krishnan, M. Panwar, and G. K. Venayagamoorthy, “Ev charging coordination via adaptive dynamic programming,” *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2083–2094, 2021.

[18] X. Shi, Y. Xu, Q. Guo, H. Sun, and W. Jiang, “Distributed optimal energy management in microgrids,” *IEEE Transactions on Smart Grid*, vol. 6, no. 3, pp. 1137–1146, 2015.

[19] P. Kotiranta, M. Junkkari, and J. Nummenmaa, “Performance of graph and relational databases in complex queries,” *Applied Sciences*, vol. 12, no. 13, 2022. [Online]. Available: https://www.mdpi.com/2076-3417/12/13/6490